

Long-term trends: Total mobile sea lice

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Mean sea lice dataset (SL_Mean)

```
SL_Mean<-read.table("MeanDataset.csv",  
  header = TRUE,sep = ',',fill = TRUE,dec = ".",na.strings = "NA")
```

```
library(ggplot2)  
library(glmmTMB)  
suppressMessages(library(lmtest))  
library(performance)  
library(gridExtra)  
library(grid)
```

Data adjustment before analysis

The outliers were identified: sites with less than 120 inspections, fish that stayed more than 24 months in the seawater, and outliers (total mobile sea lice mean values higher than 90).

```
# Excluding fish stock transported to sea water in 2023  
SL_Mean1<-SL_Mean[!(SL_Mean$New_Class %in% c("2023 S1/2","2023 S1")),]  
  
# Sites with less than 120 inspections  
Insp<-aggregate(SL_Mean1$SplDate,list(SL_Mean1$SiteName),function(x){length(unique(x))})  
Insp_Sel<-Insp[Insp$x<120,]$Group.1  
  
# Fish more than 24 months in the sea water, group '30' includes 25-30 in seawater  
SL_Mean2<-SL_Mean1[SL_Mean1$TimeSea<30,]  
SL_Mean3<-SL_Mean2[!(SL_Mean2$SiteName %in% Insp_Sel),]  
  
# Outliers  
SL_Pen<-SL_Mean3[SL_Mean3$MeanLepTotal<90,]
```

Comparison of raw and cleaned datasets

```
# Number of counts per fish  
nrow(SL_Mean)
```

```
## [1] 16955
```

```
nrow(SL_Pen)
```

```
## [1] 13484
```

```
# Number of salmon farms  
length(unique(SL_Mean$SiteName))
```

```
## [1] 67
```

```
length(unique(SL_Pen$SiteName))
```

```
## [1] 28
```

```
# Mean and sd of sea lice counts  
mean(SL_Pen$MeanLepTotal);sd(SL_Pen$MeanLepTotal)
```

```
## [1] 3.881866
```

```
## [1] 8.026194
```

The format of variables in the dataset (SL_Pen), sea lice values recorded as a mean value per fish per farm, was adjusted.

```
SL_Pen$Year_f<-as.factor(SL_Pen$Year)  
SL_Pen$TimeSea_f<-as.factor(SL_Pen$TimeSea)  
SL_Pen$Year_n<-as.numeric(SL_Pen$Year_f)/10  
SL_Pen$NewMonth<-factor(SL_Pen$NewMonth,levels=c('Jan/Dec','Feb','Mar1','Mar2',  
  'Apr1','Apr2','May1','May2','Jun','Jul','Aug','Sep','Oct','Nov'))
```

A new dataset (SL_Farm), sea lice values recorded as a mean value per fish per farm, was created.

```
SL_Farm<-aggregate(SL_Pen$MeanLepTotal,list(SL_Pen$Year,SL_Pen$Year_f,  
  SL_Pen$SiteName,SL_Pen$NewMonth,SL_Pen$Latitude,SL_Pen$Longitude,  
  SL_Pen$Month,SL_Pen$Class,SL_Pen$TimeSea_f,SL_Pen$SplDate),mean)  
colnames(SL_Farm)<-c('Year','Year_f','SiteName','NewMonth','Latitude','Longitude',  
  'Month','Class','TimeSea','Date','Lep')
```

The format of variables in the dataset (SL_Farm) was adjusted.

```
SL_Farm$TimeSea<-as.factor(SL_Farm$TimeSea)  
  
# Number of mean values per farm in the dataset  
nrow(SL_Farm)
```

```
## [1] 7400
```

Year as categorical predictor

The temporal autocorrelation for model at the pen (mean value of sea lice per fish per pen) and at the farm (mean value of sea lice per fish per farm) level were compared.

Pen level

The SL_Pen dataset (mean value per fish per pen) was used in a GLM that included all relevant predictors.

```
m1<-glmmTMB(MeanLepTotal~Class+Year_f+NewMonth+TimeSea_f+CageType+scale(Latitude)+
            scale(Longitude)+(1|SiteName),ziformula=~Class+Year_f+NewMonth+
            TimeSea_f+CageType+scale(Latitude)+scale(Longitude),
            family=ziGamma(link="log"),data=SL_Pen)
```

The temporal autocorrelation was tested.

```
dwtest(m1)
```

```
##
## Durbin-Watson test
##
## data: m1
## DW = 1.9446, p-value = 0.0005805
## alternative hypothesis: true autocorrelation is greater than 0
```

```
bgtest(m1)
```

```
##
## Breusch-Godfrey test for serial correlation of order up to 1
##
## data: m1
## LM test = 10.414, df = 1, p-value = 0.00125
```

Farm level

The SL_Farm dataset (mean value per fish per farm) was used in a GLM that included all relevant predictors.

```
m2<-glmmTMB(Lep~Year_f+NewMonth+TimeSea+Class+scale(Latitude)+
            scale(Longitude)+(1|SiteName),ziformula=~
            Year_f+NewMonth+TimeSea+Class+scale(Latitude)+scale(Longitude),
            family=ziGamma(link="log"),data=SL_Farm)
```

The model was then tested for temporal autocorrelation and the results were compared to the model (m1) at the pen level.

```
dwtest(m2)
```

```
##
## Durbin-Watson test
##
## data: m2
## DW = 1.8379, p-value = 4.047e-14
## alternative hypothesis: true autocorrelation is greater than 0
```

```
bgtest(m2)
```

```
##  
## Breusch-Godfrey test for serial correlation of order up to 1  
##  
## data: m2  
## LM test = 48.632, df = 1, p-value = 3.088e-12
```

Lower temporal autocorrelation was detected for the model at the pen level. Based on statistical diagnostics, data availability, and the research question the pen level data (SL_Pen) were used for further analysis.

The model was checked for multicollinearity.

```
check_collinearity(m1)
```

```
## # Check for Multicollinearity  
##  
## * conditional component:  
##  
## Low Correlation  
##  
##          Term  VIF    VIF 95% CI adj. VIF Tolerance Tolerance 95% CI  
##          Class 1.55 [1.51, 1.58]    1.24    0.65    [0.63, 0.66]  
##          Year_f 2.04 [1.99, 2.09]    1.01    0.49    [0.48, 0.50]  
##          NewMonth 2.12 [2.07, 2.17]    1.03    0.47    [0.46, 0.48]  
##          TimeSea_f 1.85 [1.80, 1.89]    1.11    0.54    [0.53, 0.55]  
##          CageType 1.04 [1.03, 1.06]    1.02    0.96    [0.94, 0.97]  
## scale(Latitude) 2.66 [2.59, 2.74]    1.63    0.38    [0.37, 0.39]  
## scale(Longitude) 2.66 [2.59, 2.74]    1.63    0.38    [0.37, 0.39]  
##  
## * zero inflated component:  
##  
## Low Correlation  
##  
##          Term  VIF    VIF 95% CI adj. VIF Tolerance Tolerance 95% CI  
##          Year_f 1.43 [1.40, 1.46]    1.01    0.70    [0.68, 0.72]  
##          NewMonth 1.66 [1.62, 1.70]    1.02    0.60    [0.59, 0.62]  
##          TimeSea_f 1.57 [1.53, 1.60]    1.08    0.64    [0.62, 0.65]  
## scale(Latitude) 3.09 [3.00, 3.17]    1.76    0.32    [0.32, 0.33]  
## scale(Longitude) 3.04 [2.96, 3.13]    1.74    0.33    [0.32, 0.34]
```

The predictors were selected using a backwards step-wise approach.

```
m1.1<-glmmTMB(MeanLepTotal~NewMonth+TimeSea_f+Class+CageType+scale(Latitude)+  
              scale(Longitude)+(1|SiteName),ziformula=~NewMonth+  
              TimeSea_f+Class+CageType+scale(Latitude)+scale(Longitude),  
              family=ziGamma(link="log"),data=SL_Pen)  
m1.2<-glmmTMB(MeanLepTotal~Year_f+TimeSea_f+Class+CageType+scale(Latitude)+  
              scale(Longitude)+(1|SiteName),ziformula=~Year_f+  
              TimeSea_f+Class+CageType+scale(Latitude)+scale(Longitude),  
              family=ziGamma(link="log"),data=SL_Pen)  
m1.3<-glmmTMB(MeanLepTotal~Year_f+NewMonth+Class+CageType+scale(Latitude)+
```

```

        scale(Longitude)+(1|SiteName),ziformula=~Year_f+NewMonth+
        Class+CageType+scale(Latitude)+scale(Longitude),
        family=ziGamma(link="log"),data=SL_Pen)
m1.4<-glmmTMB(MeanLepTotal~Year_f+NewMonth+TimeSea_f+CageType+scale(Latitude)+
        scale(Longitude)+(1|SiteName),ziformula=~Year_f+NewMonth+
        TimeSea_f+CageType+scale(Latitude)+scale(Longitude),
        family=ziGamma(link="log"),data=SL_Pen)
m1.5<-glmmTMB(MeanLepTotal~Year_f+NewMonth+TimeSea_f+Class+scale(Latitude)+
        scale(Longitude)+(1|SiteName),ziformula=~Year_f+NewMonth+
        TimeSea_f+Class+scale(Latitude)+scale(Longitude),
        family=ziGamma(link="log"),data=SL_Pen)
m1.6<-glmmTMB(MeanLepTotal~Year_f+NewMonth+TimeSea_f+Class+CageType+
        scale(Longitude)+(1|SiteName),ziformula=~Year_f+NewMonth+
        TimeSea_f+Class+CageType+scale(Longitude),
        family=ziGamma(link="log"),data=SL_Pen)
m1.7<-glmmTMB(MeanLepTotal~Year_f+NewMonth+TimeSea_f+Class+CageType+scale(Latitude)+
        (1|SiteName),ziformula=~Year_f+NewMonth+
        TimeSea_f+Class+CageType+scale(Latitude),
        family=ziGamma(link="log"),data=SL_Pen)

AIC(m1,m1.1,m1.2,m1.3,m1.4,m1.5,m1.6,m1.7)

```

```

##      df      AIC
## m1    104 55087.72
## m1.1   44 56543.66
## m1.2   78 56120.40
## m1.3   98 58644.14
## m1.4  102 55114.27
## m1.5  102 55085.10
## m1.6  102 55474.53
## m1.7  102 55258.70

```

```

BIC(m1,m1.1,m1.2,m1.3,m1.4,m1.5,m1.6,m1.7)

```

```

##      df      BIC
## m1    104 55868.69
## m1.1   44 56874.07
## m1.2   78 56706.12
## m1.3   98 59380.05
## m1.4  102 55880.21
## m1.5  102 55851.04
## m1.6  102 56240.48
## m1.7  102 56024.64

```

Best fit was determined for the model m1.5 without CageType. The final model was then tested for overdispersion and the explained variance was determined.

```

# Overdispersion
E2 <- resid(m1.5, type = "pearson")
N <- nrow(SL_Pen)
p <- length(coef(m1.5))
sum(E2^2) / (N - p)

```

```
## [1] 1.568947
```

```
# Explained variance  
r2_nakagawa(m1.5)
```

```
## # R2 for Mixed Models  
##  
## Conditional R2: 0.570  
## Marginal R2: 0.475
```

The results of the final model were checked.

```
summary(m1.5)
```

```
## Family: Gamma ( log )  
## Formula:  
## MeanLepTotal ~ Year_f + NewMonth + TimeSea_f + Class + scale(Latitude) +  
## scale(Longitude) + (1 | SiteName)  
## Zero inflation:  
## ~Year_f + NewMonth + TimeSea_f + Class + scale(Latitude) + scale(Longitude)  
## Data: SL_Pen  
##  
## AIC BIC logLik -2*log(L) df.resid  
## 55085.1 55851.0 -27440.6 54881.1 13382  
##  
## Random effects:  
##  
## Conditional model:  
## Groups Name Variance Std.Dev.  
## SiteName (Intercept) 0.2016 0.449  
## Number of obs: 13484, groups: SiteName, 28  
##  
## Dispersion estimate for Gamma family (sigma^2): 1.5  
##  
## Conditional model:  
## Estimate Std. Error z value Pr(>|z|)  
## (Intercept) 1.23532 0.14406 8.58 < 2e-16 ***  
## Year_f1994 -1.35173 0.12250 -11.03 < 2e-16 ***  
## Year_f1995 -1.29585 0.11952 -10.84 < 2e-16 ***  
## Year_f1996 -0.74793 0.11671 -6.41 1.47e-10 ***  
## Year_f1997 -0.71162 0.11690 -6.09 1.15e-09 ***  
## Year_f1998 -0.98130 0.11761 -8.34 < 2e-16 ***  
## Year_f1999 -0.62599 0.11607 -5.39 6.93e-08 ***  
## Year_f2000 -1.38175 0.11743 -11.77 < 2e-16 ***  
## Year_f2001 -1.58343 0.11790 -13.43 < 2e-16 ***  
## Year_f2002 -1.17582 0.11704 -10.05 < 2e-16 ***  
## Year_f2003 -0.88372 0.11807 -7.48 7.18e-14 ***  
## Year_f2004 -0.88559 0.12070 -7.34 2.19e-13 ***  
## Year_f2005 -0.75034 0.12115 -6.19 5.89e-10 ***  
## Year_f2006 -0.45577 0.12003 -3.80 0.000146 ***  
## Year_f2007 -0.37172 0.12161 -3.06 0.002237 **  
## Year_f2008 -0.56456 0.12328 -4.58 4.66e-06 ***  
## Year_f2009 -0.91560 0.12413 -7.38 1.63e-13 ***
```

```

## Year_f2010      -0.76553    0.13233   -5.78 7.25e-09 ***
## Year_f2011      -1.46351    0.13068  -11.20 < 2e-16 ***
## Year_f2012      -0.82521    0.12678   -6.51 7.58e-11 ***
## Year_f2013      -1.66913    0.13174  -12.67 < 2e-16 ***
## Year_f2014      -1.14496    0.12547   -9.13 < 2e-16 ***
## Year_f2015      -1.10152    0.12398   -8.88 < 2e-16 ***
## Year_f2016      -1.47364    0.12708  -11.60 < 2e-16 ***
## Year_f2017      -1.04116    0.12417   -8.39 < 2e-16 ***
## Year_f2018      -1.65489    0.13048  -12.68 < 2e-16 ***
## Year_f2019      -0.73335    0.12732   -5.76 8.41e-09 ***
## Year_f2020      -1.24329    0.12542   -9.91 < 2e-16 ***
## Year_f2021      -1.01844    0.12907   -7.89 3.01e-15 ***
## Year_f2022      -1.09802    0.12963   -8.47 < 2e-16 ***
## Year_f2023      -2.51723    0.17885  -14.07 < 2e-16 ***
## NewMonthFeb     -0.07374    0.05998   -1.23 0.218972
## NewMonthMar1    -0.21880    0.05901   -3.71 0.000209 ***
## NewMonthMar2    -0.36501    0.06087   -6.00 2.02e-09 ***
## NewMonthApr1    -0.65842    0.06205  -10.61 < 2e-16 ***
## NewMonthApr2    -0.66357    0.06278  -10.57 < 2e-16 ***
## NewMonthMay1    -0.74463    0.05927  -12.56 < 2e-16 ***
## NewMonthMay2    -0.65199    0.06142  -10.62 < 2e-16 ***
## NewMonthJun     -0.85359    0.05989  -14.25 < 2e-16 ***
## NewMonthJul     -0.65784    0.06030  -10.91 < 2e-16 ***
## NewMonthAug      0.06919    0.06146    1.13 0.260285
## NewMonthSep      0.39188    0.06268    6.25 4.06e-10 ***
## NewMonthOct      0.03563    0.06208    0.57 0.566028
## NewMonthNov      0.03944    0.06024    0.65 0.512617
## TimeSea_f12      1.01908    0.03689   27.63 < 2e-16 ***
## TimeSea_f18      1.80507    0.03450   52.32 < 2e-16 ***
## TimeSea_f24      2.19061    0.05497   39.85 < 2e-16 ***
## ClassS1/2       -0.18659    0.03414   -5.47 4.61e-08 ***
## scale(Latitude)  0.58068    0.14964    3.88 0.000104 ***
## scale(Longitude) -0.30180    0.14529   -2.08 0.037780 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Zero-inflation model:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)  -2.72367    0.45125  -6.036 1.58e-09 ***
## Year_f1994     1.79929    0.46049   3.907 9.33e-05 ***
## Year_f1995     0.86817    0.47997   1.809 0.070484 .
## Year_f1996     0.77381    0.48334   1.601 0.109386
## Year_f1997     1.01098    0.48129   2.101 0.035680 *
## Year_f1998     1.50997    0.46613   3.239 0.001198 **
## Year_f1999     0.47766    0.50118   0.953 0.340553
## Year_f2000     1.74495    0.45632   3.824 0.000131 ***
## Year_f2001     2.40945    0.45380   5.309 1.10e-07 ***
## Year_f2002     0.98464    0.46979   2.096 0.036089 *
## Year_f2003     0.16817    0.51710   0.325 0.745020
## Year_f2004     1.36743    0.47481   2.880 0.003977 **
## Year_f2005     1.40872    0.47386   2.973 0.002951 **
## Year_f2006     0.23384    0.51368   0.455 0.648940
## Year_f2007     1.01724    0.48926   2.079 0.037606 *
## Year_f2008     1.23650    0.47672   2.594 0.009493 **

```

```
## Year_f2009      1.85842    0.46876    3.965 7.35e-05 ***
## Year_f2010      2.12577    0.46748    4.547 5.43e-06 ***
## Year_f2011      2.64533    0.45864    5.768 8.03e-09 ***
## Year_f2012      2.66452    0.45715    5.828 5.59e-09 ***
## Year_f2013      2.51065    0.45872    5.473 4.42e-08 ***
## Year_f2014      2.51139    0.45446    5.526 3.27e-08 ***
## Year_f2015      1.58715    0.47234    3.360 0.000779 ***
## Year_f2016      2.14210    0.46328    4.624 3.77e-06 ***
## Year_f2017      1.14382    0.48773    2.345 0.019018 *
## Year_f2018      2.08538    0.46773    4.458 8.25e-06 ***
## Year_f2019      2.32874    0.45921    5.071 3.95e-07 ***
## Year_f2020      1.62114    0.47134    3.439 0.000583 ***
## Year_f2021      2.56711    0.45540    5.637 1.73e-08 ***
## Year_f2022      2.23002    0.46315    4.815 1.47e-06 ***
## Year_f2023      3.87126    0.50106    7.726 1.11e-14 ***
## NewMonthFeb     -0.46107    0.15925   -2.895 0.003788 **
## NewMonthMar1    -0.33116    0.15341   -2.159 0.030874 *
## NewMonthMar2    -0.65251    0.16598   -3.931 8.45e-05 ***
## NewMonthApr1     0.10934    0.15060    0.726 0.467836
## NewMonthApr2    -0.41104    0.16578   -2.480 0.013157 *
## NewMonthMay1     0.38992    0.14578    2.675 0.007480 **
## NewMonthMay2     0.35225    0.15141    2.327 0.019991 *
## NewMonthJun      0.47215    0.14256    3.312 0.000926 ***
## NewMonthJul     -0.09750    0.15116   -0.645 0.518952
## NewMonthAug     -0.48501    0.15918   -3.047 0.002312 **
## NewMonthSep     -0.76184    0.16852   -4.521 6.16e-06 ***
## NewMonthOct     -0.03831    0.17378   -0.220 0.825526
## NewMonthNov      0.20256    0.16815    1.205 0.228349
## TimeSea_f12     -1.24611    0.07812  -15.950 < 2e-16 ***
## TimeSea_f18     -2.21068    0.08738  -25.301 < 2e-16 ***
## TimeSea_f24     -3.20639    0.21996  -14.577 < 2e-16 ***
## ClassS1/2       0.05488    0.06820    0.805 0.420993
## scale(Latitude) -0.90412    0.04715  -19.176 < 2e-16 ***
## scale(Longitude) 0.69654    0.05460   12.756 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
car::Anova(m1.5)
```

```
## Analysis of Deviance Table (Type II Wald chisquare tests)
##
## Response: MeanLepTotal
##              Chisq Df Pr(>Chisq)
## Year_f      1053.9720 30 < 2.2e-16 ***
## NewMonth     920.6703 13 < 2.2e-16 ***
## TimeSea_f    3090.4658  3 < 2.2e-16 ***
## Class        29.8751  1 4.608e-08 ***
## scale(Latitude) 15.0578  1 0.0001043 ***
## scale(Longitude) 4.3149  1 0.0377795 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Visualization of results

The sea lice values for all levels of predictors with the confidential interval were predicted.

```
newdata2 <- data.frame(
  Class=factor(rep(rep(rep(rep(1:2),13),31),14),4),levels=1:2,
               labels=levels(as.factor(SL_Pen$Class))),
  Year_f=factor(rep(rep(rep(1:31,each=26),14),4),levels=1:31,
               labels=levels(SL_Pen$Year_f)),
  NewMonth=factor(rep(rep(1:14,each=806),4),levels=1:14,
                  labels=levels(SL_Pen$NewMonth)),
  TimeSea_f=factor(rep(1:4,each=11284),levels = 1:4,
                  labels =levels(SL_Pen$TimeSea_f)),
  Latitude=mean(SL_Pen$Latitude),
  Longitude=mean(SL_Pen$Longitude))

newdata2 <- cbind(newdata2, predict(m1.5, newdata2, type = "link", se.fit=TRUE,re.form=NA))

newdata2 <- within(newdata2, {
  Abun<- exp(fit)
  LL <- exp(fit - 1.96 * se.fit)
  UL <- exp(fit + 1.96 * se.fit)
})
```

Mean values of sea lice year were calculated from predicted sea lice counts.

```
A<-aggregate(newdata2$Abun,list(newdata2$Year_f),mean)
B<-aggregate(newdata2$LL,list(newdata2$Year_f),mean)
C<-aggregate(newdata2$UL,list(newdata2$Year_f),mean)
colnames(A)<-c('Year','Abun')
colnames(B)<-c('Year','LL')
colnames(C)<-c('Year','UL')
AA<-merge(A,B, by="Year")
BB<-merge(AA,C, by="Year")
```

The final data were visualized together with the raw data for every year at the level of farm

```
BB$Year<-as.numeric(1993:2023)

# Predicted sea lice counts
A<-ggplot(BB, aes(Year, Abun)) +
  geom_ribbon(aes(ymin = LL, ymax = UL), colour = 'grey80', fill = "grey80") +
  geom_point() +
  geom_line(linewidth = 1.5) +
  ylab('[annual mean per fish per pen]')+
  theme_minimal(base_size = 10) +
  theme(legend.position = "none",
        panel.grid.minor = element_blank()) +
  scale_x_continuous(name = "Year",limits = c(1992.5, 2023.5),
                     breaks = seq(1993, 2023, 5)) +
  coord_cartesian(xlim = c(1993, 2023))

# Observed sea lice counts
```

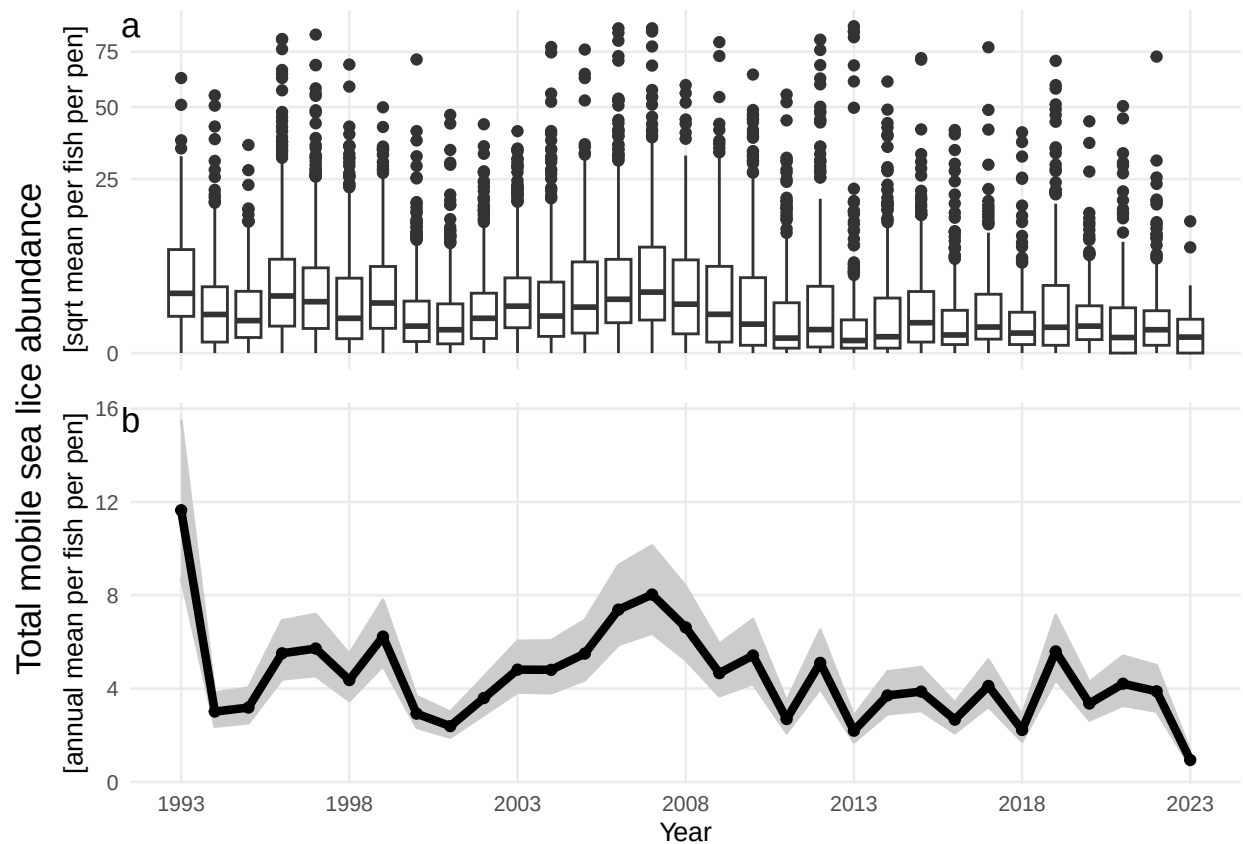
```

SL_Pen$Year <- as.numeric(as.character(SL_Pen$Year_f))
B<-ggplot(SL_Pen, aes(x = Year, y = MeanLepTotal, group = Year)) +
  geom_boxplot() +
  scale_y_sqrt() +
  ylab('[sqrt mean per fish per pen]')+
  theme_minimal(base_size = 10) +
  theme(legend.position = "none",
        axis.title.x = element_blank(),
        axis.text.x = element_blank(),
        panel.grid.minor = element_blank()) +
  scale_x_continuous(limits = c(1992.5, 2023.5),
                     breaks = seq(1993, 2023, 5)) +
  coord_cartesian(xlim = c(1993, 2023))

grid.arrange(B,A,ncol = 1,
             heights = c(1, 1.2),left = textGrob("Total mobile sea lice abundance",
             gp = gpar(fontsize = 13),rot = 90))

# Adding labels A and B
grid.text("a", x = unit(0.10, "npc"), y = unit(.968, "npc"),
         gp = gpar(fontsize = 13, col = "black"))
grid.text("b", x = unit(0.10, "npc"), y = unit(.51, "npc"),
         gp = gpar(fontsize = 13, col = "black"))

```



The predicted values were used to calculate mean values per month to visualize the seasonal changes in sea lice abundance.

```

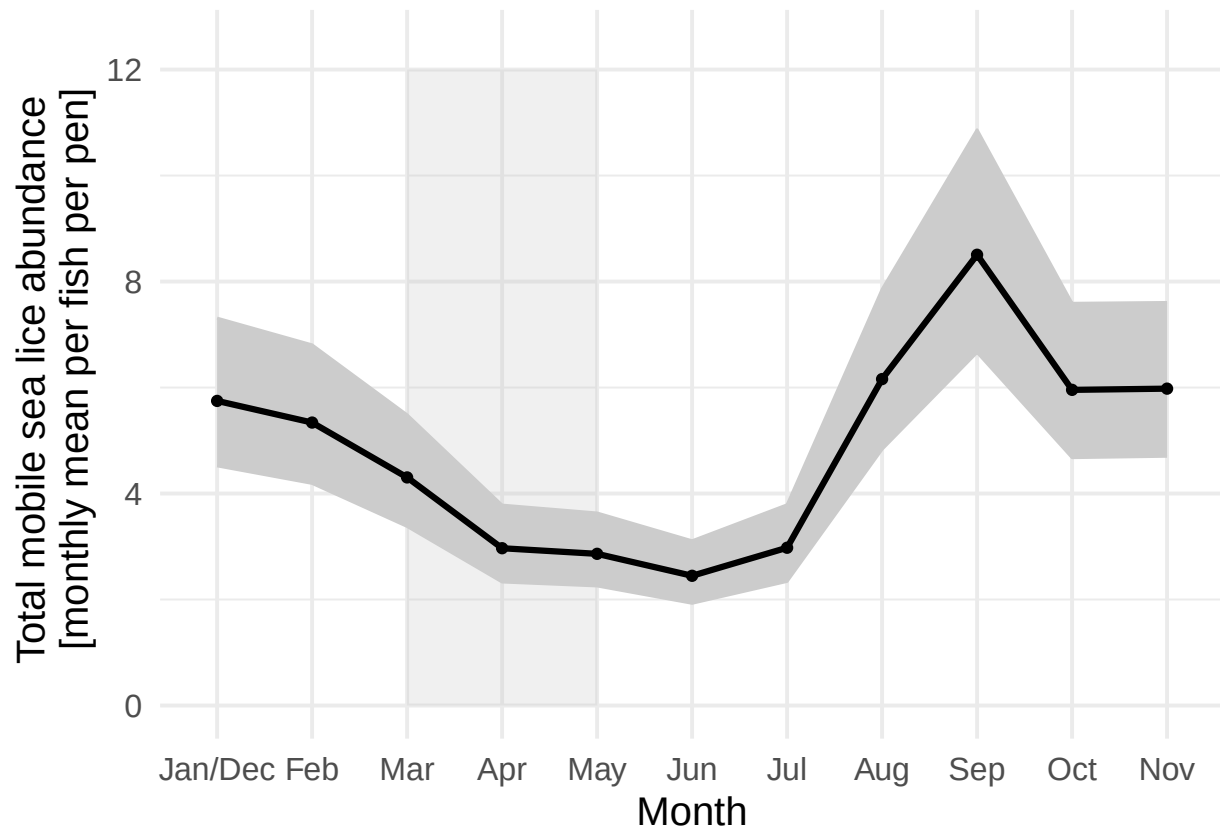
newdata2$NewMonth_n<-substr(as.character(newdata2$NewMonth),1,3)
newdata2[newdata2$NewMonth_n=='Jan',]$NewMonth_n<- 'Jan/Dec'

A<-aggregate(newdata2$Abun,list(newdata2$NewMonth_n),mean)
B<-aggregate(newdata2$LL,list(newdata2$NewMonth_n),mean)
C<-aggregate(newdata2$UL,list(newdata2$NewMonth_n),mean)
colnames(A)<-c('Month','Abun')
colnames(B)<-c('Month','LL')
colnames(C)<-c('Month','UL')
AA<-merge(A,B, by="Month")
BB<-merge(AA,C, by="Month")

# Common x-axis labels
BB$Month<-factor(BB$Month,levels=c("Jan/Dec","Feb","Mar",
  "Apr","May","Jun","Jul","Aug","Sep","Oct","Nov"))
BLUE <- data.frame(X=c('Mar','Apr','May'), Y = 12)

# Predicted sea lice counts
ggplot(data=BB,aes(Month,Abun,group=1))+
  geom_area(aes(x=X,y=Y),data=BLUE,alpha=0.3,fill="grey80") +
  geom_ribbon(aes(ymin = LL, ymax=UL),colour = 'grey80',fill="grey80") +
  geom_point() +
  geom_line(linewidth = 1.1)+
  theme_minimal(base_size = 15) +
  ylab('Total mobile sea lice abundance \n [monthly mean per fish per pen]')+
  xlab('Month')+
  ylim(0,12.5)

```



Year as numeric

Pen level

The dataset at the pen level was used to fit a model including all predictors.

```
m3<-glmmTMB(MeanLepTotal~Year+NewMonth+TimeSea_f+Class+CageType+scale(Latitude)+
  scale(Longitude)+(1|SiteName),ziformula=~Year+NewMonth+
  TimeSea_f+Class+CageType+scale(Latitude)+scale(Longitude),
  family=ziGamma(link="log"),data=SL_Pen)
```

The temporal autocorrelation was determined.

```
dwtest(m3)
```

```
##
## Durbin-Watson test
##
## data: m3
## DW = 1.9417, p-value = 0.0003231
## alternative hypothesis: true autocorrelation is greater than 0
```

```
bgtest(m3)
```

```
##  
## Breusch-Godfrey test for serial correlation of order up to 1  
##  
## data: m3  
## LM test = 11.5, df = 1, p-value = 0.0006959
```

Farm level

The dataset at the farm level was used to fit a model including all predictors.

```
m4<-glmmTMB(Lep~Year+NewMonth+TimeSea+Class+scale(Latitude)+  
            scale(Longitude)+(1|SiteName),ziformula=~Year+  
            NewMonth+TimeSea+Class+scale(Latitude)+scale(Longitude),  
            family=ziGamma(link="log"),data=SL_Farm)
```

The temporal autocorrelation was determined and compared to the model at the pen level.

```
dwtest(m4)
```

```
##  
## Durbin-Watson test  
##  
## data: m4  
## DW = 1.7698, p-value < 2.2e-16  
## alternative hypothesis: true autocorrelation is greater than 0
```

```
bgtest(m4)
```

```
##  
## Breusch-Godfrey test for serial correlation of order up to 1  
##  
## data: m4  
## LM test = 98.096, df = 1, p-value < 2.2e-16
```

Lower temporal autocorrelation was detected for the model at the pen level. Based on statistical diagnostics, data availability, and the research question the pen level data (SL_Pen) were used for further analysis.

```
check_collinearity(m3)
```

```
## # Check for Multicollinearity  
##  
## * conditional component:  
##  
## Low Correlation  
##  
##          Term  VIF    VIF 95% CI adj. VIF Tolerance Tolerance 95% CI  
##          Year 1.38 [1.35, 1.41]    1.17      0.72      [0.71, 0.74]
```

```
##           NewMonth 1.74 [1.70, 1.78]      1.02      0.57      [0.56, 0.59]
##           TimeSea_f 1.70 [1.66, 1.74]      1.09      0.59      [0.58, 0.60]
##           Class 1.39 [1.36, 1.42]      1.18      0.72      [0.70, 0.73]
##           CageType 1.02 [1.01, 1.05]      1.01      0.98      [0.95, 0.99]
##   scale(Latitude) 2.66 [2.59, 2.73]      1.63      0.38      [0.37, 0.39]
##   scale(Longitude) 2.66 [2.59, 2.73]      1.63      0.38      [0.37, 0.39]
##
## * zero inflated component:
##
## Low Correlation
##
##           Term VIF      VIF 95% CI adj. VIF Tolerance Tolerance 95% CI
##           Year 1.13 [1.11, 1.16]      1.06      0.88      [0.87, 0.90]
##           NewMonth 1.57 [1.54, 1.61]      1.02      0.64      [0.62, 0.65]
##           TimeSea_f 1.44 [1.41, 1.47]      1.06      0.70      [0.68, 0.71]
##   scale(Latitude) 3.07 [2.98, 3.16]      1.75      0.33      [0.32, 0.34]
##   scale(Longitude) 3.03 [2.95, 3.12]      1.74      0.33      [0.32, 0.34]
```

The predictors were selected using a backwards step-wise approach.

```
m3.1<-glmmTMB(MeanLepTotal~NewMonth+TimeSea_f+Class+CageType+scale(Latitude)+
  scale(Longitude)+(1|SiteName),ziformula=~NewMonth+
  TimeSea_f+Class+CageType+scale(Latitude)+scale(Longitude),
  family=ziGamma(link="log"),data=SL_Pen)
m3.2<-glmmTMB(MeanLepTotal~Year+TimeSea_f+Class+CageType+scale(Latitude)+
  scale(Longitude)+(1|SiteName),ziformula=~Year+
  TimeSea_f+Class+CageType+scale(Latitude)+scale(Longitude),
  family=ziGamma(link="log"),data=SL_Pen)
m3.3<-glmmTMB(MeanLepTotal~Year+NewMonth+Class+CageType+scale(Latitude)+
  scale(Longitude)+(1|SiteName),ziformula=~Year+NewMonth+
  Class+CageType+scale(Latitude)+scale(Longitude),
  family=ziGamma(link="log"),data=SL_Pen)
m3.4<-glmmTMB(MeanLepTotal~Year+NewMonth+TimeSea_f+CageType+scale(Latitude)+
  scale(Longitude)+(1|SiteName),ziformula=~Year+NewMonth+
  TimeSea_f+CageType+scale(Latitude)+scale(Longitude),
  family=ziGamma(link="log"),data=SL_Pen)
m3.5<-glmmTMB(MeanLepTotal~Year+NewMonth+TimeSea_f+Class+scale(Latitude)+
  scale(Longitude)+(1|SiteName),ziformula=~Year+NewMonth+
  TimeSea_f+Class+scale(Latitude)+scale(Longitude),
  family=ziGamma(link="log"),data=SL_Pen)
m3.6<-glmmTMB(MeanLepTotal~Year+NewMonth+TimeSea_f+Class+CageType+
  scale(Longitude)+(1|SiteName),ziformula=~Year+NewMonth+
  TimeSea_f+Class+CageType+scale(Longitude),
  family=ziGamma(link="log"),data=SL_Pen)
m3.7<-glmmTMB(MeanLepTotal~Year+NewMonth+TimeSea_f+Class+CageType+scale(Latitude)+
  (1|SiteName),ziformula=~Year+NewMonth+
  TimeSea_f+Class+CageType+scale(Latitude),
  family=ziGamma(link="log"),data=SL_Pen)

AIC(m3,m3.1,m3.2,m3.3,m3.4,m3.5,m3.6,m3.7)
```

```
##      df      AIC
## m3    46 56313.87
```

```
## m3.1 44 56543.66
## m3.2 20 57190.61
## m3.3 40 59745.76
## m3.4 44 56331.44
## m3.5 44 56316.05
## m3.6 44 56735.76
## m3.7 44 56492.36
```

```
BIC(m3,m3.1,m3.2,m3.3,m3.4,m3.5,m3.6,m3.7)
```

```
##      df      BIC
## m3    46 56659.29
## m3.1  44 56874.07
## m3.2  20 57340.79
## m3.3  40 60046.13
## m3.4  44 56661.84
## m3.5  44 56646.46
## m3.6  44 57066.17
## m3.7  44 56822.77
```

Best fit was determined for the full model m3 including all predictors. The final model was then tested for overdispersion and the explained variance was determined.

```
# Overdispersion
E2 <- resid(m3, type = "pearson")
N <- nrow(SL_Pen)
p <- length(coef(m3))
sum(E2^2) / (N - p)
```

```
## [1] 1.673753
```

```
# Explained variance
r2_nakagawa(m3)
```

```
## # R2 for Mixed Models
##
## Conditional R2: 0.523
## Marginal R2: 0.420
```

The results of the final model were checked.

```
summary(m3)
```

```
## Family: Gamma ( log )
## Formula:
## MeanLepTotal ~ Year + NewMonth + TimeSea_f + Class + CageType +
## scale(Latitude) + scale(Longitude) + (1 | SiteName)
## Zero inflation:
## ~Year + NewMonth + TimeSea_f + Class + CageType + scale(Latitude) +
## scale(Longitude)
## Data: SL_Pen
```

```

##
##      AIC      BIC    logLik -2*log(L)  df.resid
##  56313.9  56659.3 -28110.9   56221.9    13438
##
## Random effects:
##
## Conditional model:
##   Groups   Name      Variance Std.Dev.
## SiteName (Intercept) 0.2048   0.4525
## Number of obs: 13484, groups: SiteName, 28
##
## Dispersion estimate for Gamma family (sigma^2): 1.58
##
## Conditional model:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)    24.758442   3.572581   6.93 4.20e-12 ***
## Year           -0.012222   0.001782  -6.86 7.00e-12 ***
## NewMonthFeb     -0.076600   0.061285  -1.25  0.21134
## NewMonthMar1    -0.172598   0.060503  -2.85  0.00433 **
## NewMonthMar2    -0.297845   0.062156  -4.79 1.65e-06 ***
## NewMonthApr1    -0.539955   0.063582  -8.49 < 2e-16 ***
## NewMonthApr2    -0.573940   0.064309  -8.92 < 2e-16 ***
## NewMonthMay1    -0.674231   0.060450 -11.15 < 2e-16 ***
## NewMonthMay2    -0.585005   0.062929  -9.30 < 2e-16 ***
## NewMonthJun     -0.789157   0.061041 -12.93 < 2e-16 ***
## NewMonthJul     -0.612600   0.061651  -9.94 < 2e-16 ***
## NewMonthAug      0.076456   0.062761   1.22  0.22314
## NewMonthSep      0.408756   0.064068   6.38 1.77e-10 ***
## NewMonthOct      0.024241   0.063368   0.38  0.70206
## NewMonthNov      0.032311   0.061610   0.52  0.59997
## TimeSea_f12      0.977252   0.037833  25.83 < 2e-16 ***
## TimeSea_f18      1.772359   0.035172  50.39 < 2e-16 ***
## TimeSea_f24      2.194082   0.055900  39.25 < 2e-16 ***
## ClassS1/2       -0.157318   0.033806  -4.65 3.26e-06 ***
## CageTypeStd      0.053937   0.023595   2.29  0.02226 *
## scale(Latitude)  0.626693   0.150756   4.16 3.22e-05 ***
## scale(Longitude) -0.345130   0.146325  -2.36  0.01834 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Zero-inflation model:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)   -99.673540   7.259157 -13.731 < 2e-16 ***
## Year           0.049162   0.003615  13.598 < 2e-16 ***
## NewMonthFeb    -0.430814   0.155503  -2.770  0.00560 **
## NewMonthMar1   -0.355000   0.149727  -2.371  0.01774 *
## NewMonthMar2   -0.647639   0.162084  -3.996 6.45e-05 ***
## NewMonthApr1    0.095717   0.147288   0.650  0.51578
## NewMonthApr2   -0.410749   0.162324  -2.530  0.01139 *
## NewMonthMay1    0.368864   0.141914   2.599  0.00934 **
## NewMonthMay2    0.342579   0.147320   2.325  0.02005 *
## NewMonthJun     0.434764   0.138329   3.143  0.00167 **
## NewMonthJul    -0.146719   0.147726  -0.993  0.32062
## NewMonthAug    -0.479587   0.155790  -3.078  0.00208 **

```

```
## NewMonthSep      -0.748933    0.165257   -4.532 5.84e-06 ***
## NewMonthOct      -0.059932    0.169855   -0.353 0.72421
## NewMonthNov       0.168225    0.164201    1.025 0.30560
## TimeSea_f12      -1.197844    0.075338  -15.900 < 2e-16 ***
## TimeSea_f18      -2.123141    0.083258  -25.501 < 2e-16 ***
## TimeSea_f24      -3.031775    0.215561  -14.065 < 2e-16 ***
## ClassS1/2         0.004578    0.065839    0.070 0.94456
## CageTypeStd      -0.057741    0.059308   -0.974 0.33026
## scale(Latitude)  -0.914021    0.045709  -19.996 < 2e-16 ***
## scale(Longitude)  0.686554    0.052953   12.965 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
car::Anova(m3)
```

```
## Analysis of Deviance Table (Type II Wald chisquare tests)
##
## Response: MeanLepTotal
##              Chisq Df Pr(>Chisq)
## Year          47.0273  1 7.001e-12 ***
## NewMonth      764.4460 13 < 2.2e-16 ***
## TimeSea_f    2921.2992  3 < 2.2e-16 ***
## Class         21.6561  1 3.262e-06 ***
## CageType       5.2254  1 0.02226 *
## scale(Latitude) 17.2808  1 3.224e-05 ***
## scale(Longitude) 5.5633  1 0.01834 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```